



MLPC 2026

Data Classification: Evaluation

Team Temporal Frameworks
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Evaluation metrics - which ones make sense?

- **ACCURACY** is misleading:
Strong class imbalance means a classifier always predicting “no event” already has high ACC.
- **F1 SCORE** - balancing **PRECISION** and **RECALL**:
Penalises both false positives and false negatives, which matters equally for rare events.
- **MACRO-AVERAGING** - all classes are equally important:
Computes F1 per class, then averages - rare classes (eg. light_switch) are not drowned out by frequent ones. Works better here than micro-F1 or weighted-F1.

$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^C F1_i$$



Baseline vs. best performance

- **TRIVIAL BASELINE** - always predict “NO EVENT”:
Yields zero true positives for every class → Macro F1 = 0
- **PERFECT SCORE (F1 = 1.0) is not realistic:**
Annotator disagreement and ambiguous segment boundaries set a ceiling below 1.0
- **Experiments confirmed the METRIC WORKS:**
Gap between baseline (0) and trained classifiers (~0.5) shows the metric is sensitive and discriminative.



Realistic scores in Task 4

- **OUR CLASSIFIERS:**
XGBoost reached Macro-F1 = 0.476, ANN with focal loss slightly better at 0.511
- **DISTINCTIVE classes score highest:**
running_water (~0.76), keyboard_typing (~0.70), vacuum_cleaner (~0.68–0.71) have sustained, unique spectral signatures
- **Brief TRANSIENT SOUNDS are hardest:**
light_switch (0.14–0.19), wardrobe/drawer (0.22–0.23), window_open_close (0.20–0.23) are short and acoustically ambiguous



Ideas to break the barrier

- **CNN on raw spectrograms:**
Learn local time-frequency patterns directly from mel spectrograms instead of hand-crafted features, might capture cues that aggregated statistics discard.
- **RNN / LSTM:**
Process segments sequentially, capturing temporal context across time (e.g. onset → sustain → offset patterns) that per-segment classifiers miss entirely.
- **Pretrained audio transformer (PANNs, BEATs):**
Fine-tune a model already trained on millions of audio samples - strong representations out-of-the-box, especially for rare and subtle sound classes.